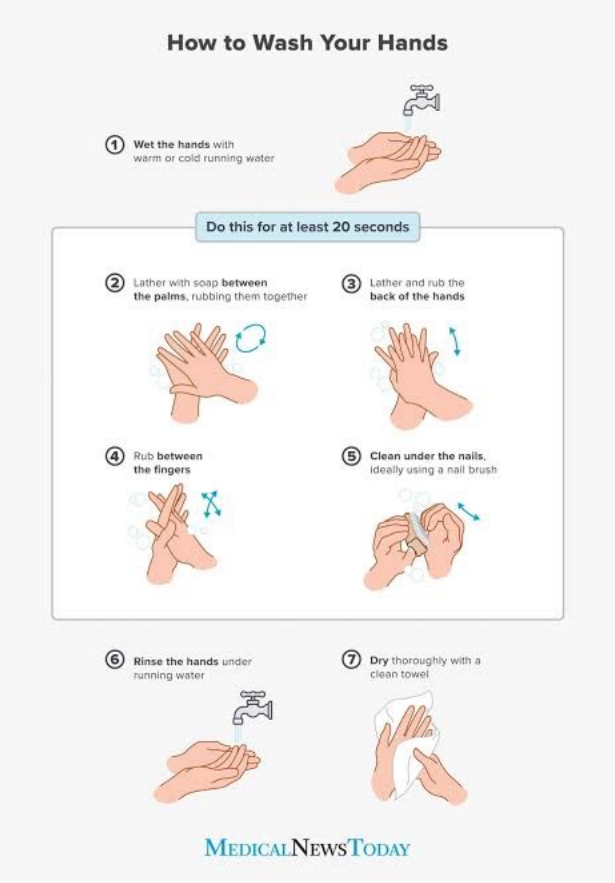


System Design Methods Methodology



*The greatest invention of the nineteenth century was the invention of the **method of invention**.*

A. N. Whitehead

Writing up Thesis F26

Which questions did you come with?

Post in chat

True or false?

1. Methods chapter can be omitted in your thesis.
2. M-chapter should describe what previous researchers did.
3. Terms *Methods* and *Methodology* are used interchangeably .
4. Never use ***we*** when describing your methods or methodology.
5. Use Past Simple to describe your methods.
6. Use Present Simple to describe your approach.

Agenda and learning outcomes

1. Purpose
- explain why describe your methods and methodology

2. Content and sequence
- **choose content for your M-chapter and sequence it**

1 

Why describe what you did?

Why describe your approach?

Approach/Method/Methodology/Design/ Experiments/Method Description and Validation

Method - how the problem was practically solved, a tool.

Methodology – the general strategy/approach to address research question/problem.

The (1) should impact which (2) are selected for research.

Once upon a time ….

There has long been this problem (or recently emerged).

This is why it is important.

Previously is has been addressed in such and such ways, which had their benefits, but also limitations.

or

No one has ever addressed this problem.

or

No one has ever approached this problem with this method.

or

No one has ever used this method on this population.

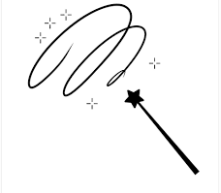
So, we decided to give it a try. **Particularly, we addressed these RQs (or tested these Hypotheses): …**

To answer those research questions/to test those hypotheses, this is what we did. And this is why we did this.

And this is what we found.

This is how our results are better or worse than previous ones. And this is why.

To conclude, our point is that …



… graduated and lived happily ever after.

Tell your reader:

- what you did in your project
- how exactly you did it, in details so that it is **replicable**
- why you chose to do it in this specific way/ways

1 

Respond in the chat:

- Why describe methods?
- Why describe methodology?
- What happens if there is no such descriptions in your thesis?

2 minutes

2 

- content
- sequence

Thesis guidelines:

4.5 Design and Methodology

In this chapter, students should describe the general structure of the study: methods, approaches or processes followed during the study. The methodology should be linked back to the aims of the thesis and the literature to explain why you are using certain methods. In the case of a system development, students should describe the Design of the system: describe each component and how they will interact to each other. They should also describe the different experiments carried out. The design and solution should be elegant and robust – consistent with industrial standards.

This chapter should answer:

- Why did you use this technology/method?
- How does the theory relate to your implementation?
- What are your underlying assumptions?
- What did you neglect and what simplifications have you made?
- What tools and methods did you use?
- Why use these tools and methods?

Typical complaints of referees:

The methods are not adequately described and are incomplete.

No data treatment is shown (statistics, replicates, etc.). Statistical analysis must be reported.

Some of the procedures used were in no way obvious. The authors should justify their rationale for choosing such procedures. At other times the authors repeated a lot of well known published data, when they could have simply used a reference.